

Appendix A. Monitor Rollback Algorithms

This Appendix details the algorithms BenMAP-CE calculates when you perform a “monitor rollback.” The monitor rollback adjusts the air quality monitoring data to reflect hypothetical changes in air pollution in a given location. The program allows you to roll back the monitoring data using three approaches: Percentage, Increment, and Rollback to Standard.

Once a set of monitors has been selected, the user may define one or more non-overlapping rollback **regions**. A region is simply an area in which you perform a specified rollback. Three rollback **types** are available:

- Percentage Rollback. Monitor values are reduced the same percentage.
- Incremental Rollback. Monitor values are changed by the same fixed increment.
- Rollback to a Standard. Monitor values are reduced so that attainment of a specified standard is reached.

Each of these rollback types has different rollback parameters associated with it.

A.1 Percentage Rollback

Percentage Rollback involves setting only two parameters - a **percentage** and a **background level**. The rollback procedure is similarly straightforward - each observation at each monitor in the region has the portion of its value which is above **background level** reduced by **percentage**.

Example: Background Level: 35; Percentage: 25

Initial Observations at a monitor in rollback region:

20 20 25 59 35 51 83 35 30 67 87 79 63 35 35

If we select the background level of 35, we first calculate the portion of each observation that is above background level, that is, we subtract the background level from the initial observation level. Observations below background level are given a value of 0.

Observation portions above **background level**:

0 0 0 24 0 16 48 0 0 32 52 44 28 0 0

When we apply the rollback percentage, each observation portion gets reduced by 25%. Reduced portions above background level:

0 0 0 18 0 12 36 0 0 24 39 33 21 0 0

Then, each reduced portion is added to the background level of 35. Zero values are replaced by the initial observations. Reduced Observations:

20 20 25 53 35 47 71 35 30 59 74 68 56 35 35

A.2 Incremental Rollback

Incremental Rollback similarly involves setting only two parameters - an **increment** and a **background level**. The rollback procedure is quite similar to the percentage rollback procedure - each observation at each monitor in the region has the portion of its value which is above **background level** reduced by **increment**. The reduced values are not allowed to become negative, however. That is, the values are truncated at zero.

Example: Background Level: 35; Increment: 25 Initial Observations:

20 20 25 59 35 51 83 35 30 67 87 79 63 35 35

Observation portions above background level:

0 0 0 24 0 16 48 0 0 32 52 44 28 0 0

Reduced portions above background level:

0 0 0 0 0 0 23 0 0 7 27 19 3 0 0

Reduced Observations:

20 20 25 35 35 35 58 35 30 42 62 54 38 35 35

A.3 Rollback to a Standard

Rollback to a Standard has two groups of parameters - those associated with the **Attainment Test**, which determines whether a monitor is in attainment (meets the standard), and those associated with the **Rollback Methods**, which are used to bring out of attainment monitors into attainment.

The **Attainment Test** parameters are **Metric**, **Ordinality**, and **Standard**. A monitor is considered in attainment if the *n*th highest value of the metric specified by **Metric** is at or below the value specified by **Standard**, where *n* is the value specified by **Ordinality**. For example, if **Metric** is "TwentyFourHourDailyAverage," **Ordinality** is two, and **Standard** is eighty-five, a monitor will be considered in attainment if the second highest value of TwentyFourHourDailyAverage is at or below eighty-five.

Supported metrics for pollutants with hourly observations (Ozone) include FiveHourDailyAverage, EightHourDailyAverage, TwelveHourDailyAverage, TwentyFourHourDailyAverage, OneHourDailyMax, and EightHourDailyMax. Supported metrics for pollutants with daily observations (PM_{2.5}) include TwentyFourHourDailyAverage and AnnualAverage. For Annual Average, **Ordinality** does not apply, since there is only a single metric value to work with.

The **Rollback Method** parameters are **Interday Rollback Method**, **Interday Background Level**, **Intraday Rollback Method**, and **Intraday Background Level**. These four parameters determine the rollback procedures used to bring out of attainment monitors into attainment. The **Interday Rollback Method** and **Background Level** are used to generate target values for the metric specified by the **Attainment Test**. The **Intraday Rollback Method** and **Background Level** are used to adjust hourly observations to meet the target metric values generated in the previous step. As such, the **Intraday Rollback Method** and **Background Level** are used only for pollutants with hourly observations (ozone).

A.3.1 Interday Rollback – Generating Target Metric Values

Because standards are defined on metrics, not directly on observations, the first step in rolling back out-of-attainment monitors is generating target metric values. There are three supported rollback methods for Interday Rollbacks: Percentage, Incremental, and Peak Shaving. Each of these rollback methods requires some preprocessing of the initial monitor metric values. We will discuss this preprocessing first, and then go through Percentage, Incremental, and Peak Shaving rollbacks in turn.

The Interday Background Level specifies the portion of each metric value which cannot be affected by human intervention - we call this portion the non-anthropogenic portion. Whatever portion is left over after subtracting out the background level is referred to as the anthropogenic portion. The anthropogenic portion of the initial monitor metric values is the only part which will be affected by the Interday Rollback Method.

BenMAP calculates an out of attainment value by determining the particular monitor metric value which caused the monitor to be out of attainment - this value is the n th highest value of the metric specified by the Attainment Test metric, where n is the Attainment Test ordinality. BenMAP then calculates an anthropogenic out of attainment value by subtracting the Interday Background Level from the out of attainment value. BenMAP also calculates an anthropogenic standard by subtracting the Interday Background Level from the Attainment Test standard. Finally, BenMAP calculates a set of anthropogenic metric values and a set of non-anthropogenic metric values using the following procedure on each initial monitor metric value:

IF the metric value is less than or equal to the Interday Background Level,

non-anthropogenic metric value = metric value

anthropogenic metric value = 0

ELSE

non-anthropogenic metric value = Interday Background Level

anthropogenic metric value = metric value - Interday Background Level

A.3.1.1 Interday Rollback – Percentage

To generate target metric values using Percentage rollback, BenMAP calculates the percentage required to reduce the anthropogenic out of attainment value to exactly meet the anthropogenic standard. This percentage reduction is then applied to all of the anthropogenic metric values. Finally, these reduced anthropogenic metric values are added to the non-anthropogenic metric values to give the final target metric values.

Example:

Initial Metric Values:

30 35 50 100 80 44 67 88 90 70 50 30 55 90 80 85

Attainment Test: Highest value of metric ≤ 70

Interday Background Level: 40

Out of Attainment Value: 100

Anthropogenic Out of Attainment Value: 60 (= 100 - 40)

Anthropogenic Standard: 30 (= 70 - 40)

Percentage Reduction Required: 50% $(= (60 - 30) / 60)$

Non-Anthropogenic Metric Values:

30 35 40 40 40 40 40 40 40 40 40 30 40 40 40 40

Anthropogenic Metric Values:

0 0 10 60 40 4 27 48 50 30 10 0 15 50 40 45

Reduced Anthropogenic Metric Values:

0 0 5 30 20 2 14 24 25 15 5 0 8 25 20 23

Target Metric Values:

30 35 45 70 60 42 54 64 65 55 45 30 48 65 60 63

A.3.1.2 Interday Rollback – Incremental

To generate target metric values using Incremental Rollback, BenMAP calculates the increment required to reduce the anthropogenic out of attainment value to exactly the anthropogenic standard. This incremental reduction is then applied to all of the anthropogenic metric values (but they are not allowed to fall below zero). Finally, these